



Python for Fixed Income

COURSE 3 · MARKET DATA AND TIME SERIES

JSON and the FRED API

WEEK 1 · DAY 2

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

From one cell to three functions

`fred_params`

Builds the request

no network

`parse_fred_observations`

Reads the answer

no network

`fred_series`

Sends it, then parses

the one network function

Two of the three need no network, so they are tested on a sample file, with no key, in a fraction of a second.

Your key never shows: not in a printed URL, an error message, or a traceback.

Week 1: getting the data in

Day 1 APIs, keys, and secrets

Day 2 **JSON and the FRED API**

Day 3 Market calendars, cleaning

Day 4 Resampling, the as-of value

Day 5 Case study: the dataset

JSON in one slide

```
{
  "series_id": "DGS10",
  "complete": true,
  "notes": null,
  "observations": [
    {"date": "2026-09-30", "value": "5.29"}
  ]
}
```

Two containers: an object { } of name and value pairs, and a list [] of values in order. Plain text, so any program can read it.

JSON	Python
object { }	dict
list []	list
"text"	str
5 or 5.29	int or float
true, false	True, False
null	None

FRED sends every value as text: "5.29", not 5.29.

FRED's answer

Name	In the sample
realtime_start, _end	2026-10-02 (not used)
observation_start, _end	2026-08-03 to 2026-09-30
units	lin: levels, as published
count	43
limit	100,000 per answer
observations	a list of 43

The sample file

FRED's layout, Treasury's values, built for the course. Nothing in it was downloaded from FRED. No key, no network.

FRED Help: "Before the popularization of the World Wide Web, the data were provided in list form on a dial-in, electronic bulletin board system."

FRED Help, What is FRED? (read 2026-10-04)

A dot is missing, never zero

Date	Value	NaN kept: change	Dot as 0: change
2026-09-04	4.78		
2026-09-07	". "	(dropped)	-478 bp
2026-09-08	4.80	2 bp, from Friday	+480 bp

4.8367%

mean of 42 days, NaN left out

4.7242%

mean of 43 days, dot as 0

11.2 bp

moved by one zero

A dot becomes NaN, never a number. NaN rows are dropped, in a step you can see, before any change or window.

fred_params and parse_fred_observations

fred_params: pure, sends nothing

```
md.fred_params("DGS10", FAKE_KEY,  
               start="2026-08-03")  
# holds the key: never printed
```

parse_fred_observations

```
float("nan") if obs["value"] == "."  
else float(obs["value"])  
# never 0
```

43

rows, one NaN, float64

42

after dropna, a visible step

0.0

largest gap to the Treasury file

7

tests passed

No observations in the answer (one of FRED's error answers): ValueError, naming the keys it found.

fred_series: errors that hide the key

`response.raise_for_status()` says, with the fake key:

```
400 Client Error: Bad Request for url: ...  
...?series_id=DGS10&api_key=fakefakefakefakefakefakefake...
```

- 1 Never `raise_for_status`
- 2 Its own error: status and FRED's message
- 3 Connection error: the class name only
- 4 Raised from None: no chained traceback
- 5 Empty answer, status 200: stop

Tests with monkeypatch and a stand-in answer: no fake key and no `api_key=` anywhere in the printed error. 9 passed.

Live: the 10-year and the 2-year from 2015

```
for live_id, live_column in [("DGS10", "10 Yr"),  
                             ("DGS2", "2 Yr")]:  
    live_series = md.fred_series(live_id, fred_key, ...)
```

1. Equal on every common date (1e-9 percent)
2. Every dot on a date the file has no row for
3. No FRED number on a date the file lacks

A live cell, two ways

No key (the saved notebook):

Live cell skipped: no FRED key was found.

With your key:

one line per series, largest gap 0

Checked 2026-10-04: the Treasury file ends 2026-10-02, FRED's last DGS10 was 2026-10-01. Only common dates are compared.

The rate limit and the notice

120

requests a minute; then HTTP 429

3

requests in today's live cells

11

for the curve, one tenor at a time

Day 5 adds a cache: a series already pulled today is read from a file instead of asked for again.

This product uses the FRED® API but is not endorsed or certified by the Federal Reserve Bank of St. Louis.

FRED API Terms of Use and FRED API errors page (read 2026-10-04).

Excel side by side: Power Query reads the JSON

```
Source = Json.Document(File.Contents("...json"))
Rows = Table.FromRecords(Source[observations])
NoDots = Table.SelectRows(Rows, each [value] <> ".")
```

Query	Excel returns
Read the JSON	43 rows, value as text
Type to number	43 rows, the dot an empty cell
SelectRows first	42 rows, all numbers

Tested in Excel 16.0 build 20430, 2026-10-04, on the course's sample file.

A change across the gap

$$=(D28-D27)*100 \text{ on } 2026-09-08$$

480 bp

The empty cell counts as 0: the same wrong change as Python's dot made 0.

Excel side by side: VALUE and IFERROR

Excel side by side (tested in Excel)

`=VALUE("." )`

`=IFERROR(VALUE(B2),NA())`

`=B3-B2`

`=AVERAGE(B2:B4)`

#VALUE! like `float("." )` stopping

#N/A like errors="coerce" giving NaN

#VALUE! a number minus the text

4.23 text is skipped, COUNT 1

B2 the text ".", B3 the number 4.23, B4 the text '4.25.

Leave B2 empty and =B3-B2 returns 4.23: the empty cell counts as 0. The spreadsheet version of today's bug.

Save the day: read before you commit

git diff --cached

What the commit is about to record: the staged change, line by line.
Today every line starts with +.

```
3 files changed, 452 insertions(+)
```

git grep -n -E "[a-z0-9]{32}"

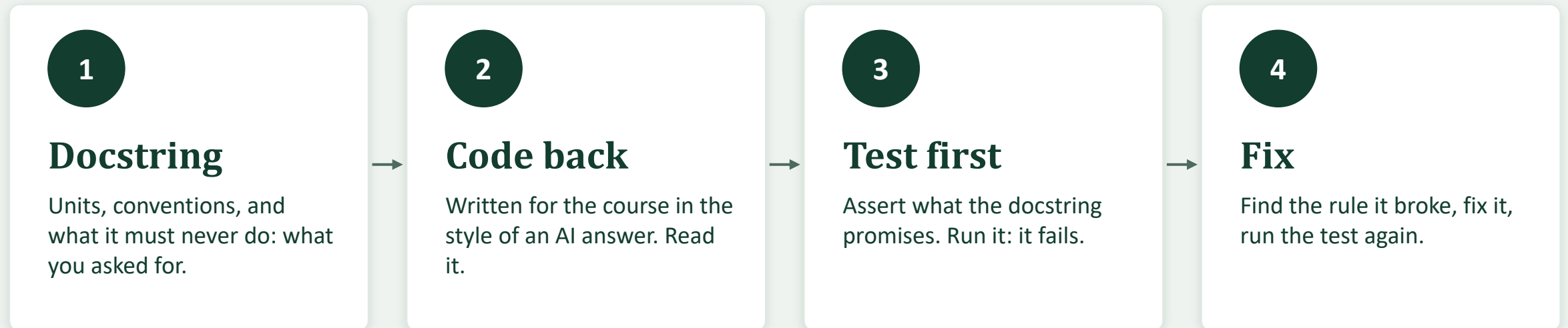
Every tracked or staged file, searched for the shape of a FRED key.
Your ignored .env is never read.

```
test_marketdata.py:12:FAKE_KEY = "fakefake...fake"
```

One hit, the course's fake key: expected. A real key there: take it out if only staged; rotate it if committed.

The AI check

The docstring is the prompt. The code that comes back runs, and has one data bug.



Today: `parse_yields` turns "." into 0.0 "so the column stays numeric": 2026-09-07 reads as a 0 percent yield.

Code that runs is not the same as code that is right.

What you can do now

- JSON: objects and lists; `json.loads` gives dictionaries and lists
- FRED's answer: metadata, count, and observations whose values are text
- A dot is NaN, never 0, dropped in a visible step
- `fred_params` and `parse_fred_observations`, tested with no network
- `fred_series` never calls `raise_for_status`; read the diff, scan for keys

Week 1, Day 3 Market calendars and cleaning: weekdays with no row, holiday calendars, and joins that invent nothing